Online Supplement

Synthetic Cultural Agents as Scoring Instruments: A Preregistered Held-Out Evaluation of Survey Item Screening

This supplement reports the two registered descriptives that the article sends here, the full tables behind several sentences of the article, the items the probe selected, and the provenance of the scoring runs. Every statistic is read from the same numbers table as the article and keeps the article's surface labels (S1 and S2 for the first and second cohorts on the strict battery, H for the retrospective held-out screen, R for the unseeded retrain, and N for the near-zero refits). Sections and tables numbered with an S belong to this supplement. As in the article, every result on surface H is retrospective, because the WVS Wave 7 file existed before the probe and is the only human source.

S1 Registered descriptives

Item-level agreement without the smallest items (H). Section 8 of the registration lists one item-level descriptive beside the primary statistic: the selected items whose signed agreement satisfies $|a_i| \geq 0.15$. The registration excludes this cut from the inference, because it conditions on the human criterion. Of the 56 items selected under the primary rule, 26 reach the cut. Their mean signed agreement is 0.115, and 69.2% of them have $a_i > 0$, against 60.7% over all 56 selected items. That is, the selected items whose human country differences correlate clearly with the GPS coordinate of their assigned dimension run in the assigned direction about two times in three. The figure describes where the agreement in A comes from and carries no interval. It is no evidence for the screen beyond the registered test reported in the article.

The face-valid reader on all of its items (H). Section 6 of the registration reports the face-valid reader's statistic on all of its assigned items, negative reciprocity included, beside the registered contrast on the imprint set. The registration counted 30 assigned items. Amendment 2 removed the one repaired item the reader had assigned (Q82), which leaves 29. Over these items the reader's A is 0.055, against 0.095 on

its 25 imprint-set items. The difference comes from its 4 negative-reciprocity items, whose A is -0.192: the human country differences on the items the reader linked to negative reciprocity by their wording run, on average, against the direction it read from their response labels. These items lie outside the imprint set and enter none of the registered contrasts.

S2 Per-dimension readings of the questionnaire readers

The registration asks for per-dimension A_d of both questionnaire readers beside the coverage-matched contrast (Table S1). The intervals come from the same registered country bootstrap as the adapters' per-dimension readings in the article, and the repaired items are removed as Amendment 2 requires. Every interval covers zero, and most cells hold a handful of items, so the table shows mainly where each reader concentrates. Both readers assign most of their imprint-set items to trust, which is why their own-item A in the article is close to their trust reading.

Table S1: Per-dimension readings of the two questionnaire readers on their own imprint-set items (H, retrospective).

	Face-valid reader		Typed decision model	
Dimension	A_d [95% interval]	Items	A_d [95% interval]	Items
Trust	0.110 [−0.154, 0.379]	19	0.151 [−0.076, 0.370]	32
Risk taking	0.044 [−0.170, 0.243]	4	−0.074 [−0.398, 0.239]	1
Altruism	0.055 [−0.163, 0.265]	2	0.049 [−0.078, 0.165]	7

Note: The typed decision model's assignment is the majority over 5 runs. Neither reader assigned an item to positive reciprocity. Intervals are percentile limits from 2,000 resamples of the 42 countries (seed 20261015), unadjusted for the number of dimensions.

S3 The second scenario bank in full

The article reports the adapter–GPS correlations of all seven strict-battery surfaces on the second bank, and those of the four imprint surfaces on the first. Table S2 adds the first-cohort values of the three flat surfaces, so that both cohorts are shown on every surface. Both cohorts use the same statistic, the same profile-block bootstrap and the same seed. Patience and negative reciprocity are flat on both banks, and their intervals are wide on both.

The article reports the bank-to-bank Pearson correlation for five surfaces. Table S3 gives both the Pearson and the Spearman coefficient for all seven, over the 26 profiles

Table S2: Adapter–GPS Spearman correlation over the 42 countries, by surface and cohort (S1 and S2, model side only).

Surface	First cohort, original bank (S1)	Second cohort, second bank (S2)
Trust	0.744 [0.500, 0.865]	0.803 [0.615, 0.876]
Risk taking	0.435 [0.085, 0.686]	0.210 [−0.199, 0.584]
Positive reciprocity	0.488 [0.120, 0.695]	0.366 [−0.068, 0.652]
Altruism	0.406 [0.020, 0.676]	0.364 [−0.057, 0.653]
Negative reciprocity	−0.028 [−0.418, 0.406]	0.239 [−0.237, 0.604]
Patience (Q43)	−0.019 [−0.414, 0.452]	0.005 [−0.403, 0.470]
Patience (Q50)	0.177 [−0.313, 0.580]	0.161 [−0.343, 0.557]

Note: Percentile intervals from a profile-block bootstrap over the 26 sign profiles the countries realize (2,000 draws, seed 20261011). In the second cohort each country takes the adapter of its sign profile.

the two banks share. The two coefficients tell the same story. Risk taking is the least stable surface across banks on either coefficient, and the two patience items, flat against the GPS, are nevertheless stable across banks. That is, a surface can be a stable property of the instrument and still carry no agreement with its anchor.

Table S3: Agreement between first-bank and second-bank surface composites over the shared sign profiles (model side only).

Surface	Pearson [95% interval]	Spearman [95% interval]
Trust	0.909 [0.797, 0.965]	0.907 [0.774, 0.957]
Risk taking	0.598 [0.229, 0.819]	0.544 [0.187, 0.772]
Positive reciprocity	0.898 [0.775, 0.960]	0.937 [0.818, 0.983]
Altruism	0.949 [0.859, 0.989]	0.944 [0.832, 0.985]
Negative reciprocity	0.708 [0.488, 0.861]	0.600 [0.230, 0.838]
Patience (Q43)	0.812 [0.699, 0.945]	0.889 [0.682, 0.971]
Patience (Q50)	0.940 [0.890, 0.975]	0.953 [0.856, 0.982]

Note: First-bank values are the means of the first-cohort country adapters within each profile, and second-bank values are the profile adapters' composites. Percentile intervals from a 2,000-draw bootstrap over profiles. No human data enter.

S4 The country-named persona route

The persona route replaces the neutral opening of the evaluation prompt with a sentence that names the country, and it is scored on the base model with no adapter. One run scored all 42 personas after a fresh base-model pass had reproduced the banked

base-model scores of the second cohort's strict run, with a largest log-probability gap of 0.0 over 140 option rows. Table S4 gives the GPS correlation of the two prompt routes by surface; the adapter columns are those of Table S2. Table S5 gives each route's correlation minus the persona route's. The planned contrast is the first-cohort difference on trust, and every other difference is descriptive. Under a country bootstrap (seed 20260905) the planned contrast is 0.716 [0.325, 1.075].

The persona route tracks none of the four imprint surfaces. Its two clear correlations are on the two patience items, and they have opposite signs.

Table S4: Prompt routes against the GPS: Spearman correlation over the 42 countries, by surface (model side only).

Surface	Sign-profile prompt	Persona (profile blocks)	Persona (country bootstrap)
Trust	0.773 [0.584, 0.844]	0.028 [−0.339, 0.381]	[−0.273, 0.330]
Risk taking	−0.448 [−0.722, −0.049]	0.195 [−0.201, 0.545]	[−0.151, 0.506]
Positive reciprocity	0.490 [0.150, 0.709]	0.033 [−0.342, 0.315]	[−0.245, 0.320]
Altruism	0.540 [0.199, 0.747]	−0.136 [−0.485, 0.214]	[−0.427, 0.197]
Negative reciprocity	0.005 [−0.440, 0.447]	0.157 [−0.140, 0.449]	[−0.153, 0.458]
Patience (Q43)	−0.195 [−0.546, 0.218]	−0.491 [−0.771, −0.124]	[−0.721, −0.214]
Patience (Q50)	0.184 [−0.299, 0.573]	0.509 [0.157, 0.730]	[0.243, 0.714]

Note: The sign-profile prompt column maps each country to its prompted profile. Profile-block intervals resample the 26 sign profiles (2,000 draws, seed 20261011); the last column resamples countries (seed 20260905) and repeats the point estimate of the column before it.

Table S5: Each route's GPS correlation minus the persona route's, by surface (model side only).

Surface	First cohort − persona	Second cohort − persona	Sign prompt − persona
Trust*	0.716 [0.191, 1.154]	0.775 [0.347, 1.144]	0.745 [0.327, 1.086]
Risk taking	0.240 [−0.156, 0.617]	0.015 [−0.436, 0.493]	−0.643 [−1.060, −0.118]
Positive reciprocity	0.454 [−0.036, 0.907]	0.332 [−0.196, 0.816]	0.457 [0.052, 0.884]
Altruism	0.543 [0.042, 0.990]	0.500 [−0.026, 0.965]	0.677 [0.198, 1.083]
Negative reciprocity	−0.185 [−0.612, 0.297]	0.082 [−0.441, 0.561]	−0.152 [−0.640, 0.374]
Patience (Q43)	0.472 [0.019, 0.986]	0.496 [0.031, 1.013]	0.295 [−0.134, 0.720]
Patience (Q50)	−0.333 [−0.840, 0.085]	−0.349 [−0.875, 0.084]	−0.326 [−0.802, 0.083]

Note: *The planned contrast, fixed before any persona was scored; every other cell is descriptive. Both correlations of a difference are recomputed on every draw of one profile-block bootstrap (2,000 draws, seed 20261011).

On the 16 countries of the development stage, rescored on the frozen stack with the first-cohort adapters, the trust correlation is 0.818 [0.554, 0.926] for the adapters and

−0.038 [−0.565, 0.480] for the persona route, with a country bootstrap. These values are descriptive. The development stage used other adapters on an earlier software stack, and its own scores enter no statistic in the article.

S5 Retraining noise by surface

The article bounds the strict-surface shifts of the unseeded retrain by their maximum. Table S6 gives the shift on each surface, in standard deviations of that surface over the 64 profiles. These shifts are the noise floor that the near-zero reading in Section S6 uses. On the recoded scale of the readout, the Pearson correlation between the retrain and the original over the 23 strict items is 0.9999.

Table S6: Unseeded retrain minus original, strict-battery surfaces, in profile standard deviations (R, model side only).

Surface	Shift
Trust	−0.025
Risk taking	−0.030
Positive reciprocity	−0.011
Altruism	−0.008
Negative reciprocity	0.033
Patience (Q43)	0.006
Patience (Q50)	−0.112

S6 The near-zero margin in full

The near-zero check refits two predeclared countries without the training pairs of every dimension whose GPS coordinate lies within 0.10 of zero, each beside a size-matched control trained on the same number of pairs drawn at random. For Great Britain (profile `P_1_1_1_1_0_1`) the rule removed altruism, negative reciprocity, positive reciprocity and risk taking, so trust and the two patience items stayed in training. For Colombia (profile `P_1_0_0_1_1_0`) it removed altruism, risk taking and trust. Table S7 gives every contrast. The predeclared reading uses only the retained imprint surfaces, which are Great Britain's trust and Colombia's positive reciprocity. The dropped dimensions are a loss-of-signal check: with no training pairs left on a dimension, its surface is expected to move away from its declared sign, so a move there bears on neither side.

The floor in the last column is the unseeded retrain's shift on the same surface (Table S6), and almost every contrast exceeds it. That floor comes from one retrain

on the full training data, and it is too small for refits trained on a fraction of the data. The size-matched controls show why: a control differs from the full-data baseline by up to 1.222 standard deviations for Great Britain and 1.499 for Colombia. Against that variation, neither retained imprint contrast stands out. We therefore claim no reversal, and the margin stays a footnote in the article.

Table S7: Near-zero refit minus its size-matched control, in profile standard deviations (N, model side only).

Country	Surface	In training	Intended	Refit − control	Floor
Great Britain	Trust*	retained	up	0.448	0.025
	Patience (Q43)	retained	up	0.078	0.006
	Patience (Q50)	retained	up	0.389	0.112
	Risk taking	dropped	up	−0.018	0.030
	Positive reciprocity	dropped	down	0.219	0.011
	Negative reciprocity	dropped	up	−0.751	0.033
	Altruism	dropped	up	0.242	0.008
Colombia	Positive reciprocity*	retained	up	−0.092	0.011
	Negative reciprocity	retained	down	−0.931	0.033
	Patience (Q43)	retained	down	−0.223	0.006
	Patience (Q50)	retained	down	−0.559	0.112
	Trust	dropped	up	−0.480	0.025
	Risk taking	dropped	down	0.086	0.030
	Altruism	dropped	up	0.062	0.008

Note: Rows marked * are the retained imprint surfaces that the predeclared reading uses. "Intended" is the direction of the unit's own sign bit. Composites are frozen-recode item means, scaled by the standard deviation of the 64 second-cohort profile composites on the same surface. The floor is the unseeded retrain's absolute shift on that surface.

S7 Items selected by the probe

Table S8 lists the 56 items the probe selected under the primary rule, by assigned dimension and sign. A plus sign means that profiles with the dimension's bit set have a higher expected response code on the item. A minus sign means a lower one. Codes follow the WVS Wave 7 questionnaire, so the substantive meaning of a direction depends on each item's response scale. The list is model-side output, frozen before any held-out human value was read. As the article reports, item-level assignments are more stable under readout variants than membership in this particular list.

Table S8: The frozen primary selection: WVS Wave 7 items by assigned dimension and sign.

Dimension	Items	Sign +	Sign −
Trust	12	Q49	Q45, Q118, Q120, Q130, Q170, Q197, Q224, Q228, Q229, Q236, Q253
Risk taking	13	Q97, Q102, Q163, Q194	Q5, Q79, Q136, Q141, Q151, Q166, Q168, Q212, Q215
Positive reciprocity	11	Q95, Q105	Q6, Q15, Q17, Q51, Q52, Q53, Q54, Q55, Q205
Altruism	20	Q96, Q122, Q125, Q135, Q138, Q198, Q227, Q231, Q241	Q27, Q124, Q128, Q150, Q160, Q161, Q162, Q209, Q213, Q216, Q242

Note: The selection file's SHA-256 hash is in Section S8. Amendment 1 repaired the wording of 12 items after a first probe pass had been computed and read, and two of them, Q153 and Q157, were in that pass's top quartile by $|t|$. Neither is in this selection, which was made on the repaired wording. Q253 is the only repaired item selected.

S8 Provenance of the scoring runs

Specifications. Each scoring condition has a pinned specification hash that fixes the prompt template, the option order, the leading space before each code and the recode. Every run reproduced its pin before a model was loaded. The strict battery's pin begins `ed3bb964b606`, and the amended held-out pool's pin begins `46447de63f7c`. The pre-amendment held-out pin, which Amendment 1 superseded, began `4d799b02`.

Code revisions. All manifests of the held-out runs (the primary readout, the three readout variants and the prompted profiles) record the checkout revision `2126045`. During a recovery, three launcher files around that checkout were patched in place on the virtual machine to fix how a child environment was built. The scoring modules are byte-identical between the patched setup and the recorded revision, which the command `git diff -quiet 2126045 HEAD - paper_b/score_runtime.py paper_b/score_plan.py paper_b/score_cohort.py` confirmed at the time. The strict-battery run of the second cohort spans two recorded revisions. All but three of its manifests record `5133387`, and the remaining three (profiles `P_0_1_1_1_1_1`, `P_1_1_1_1_1_0` and `P_1_1_1_1_1_1`) record `f34a79f`. The unseeded retrain and the strict scoring of the prompted profiles ran at `b2dbbfb`, and the near-zero refits at `d1ab226`. Between these revisions the readout path changed in four ways, none of which touches prompt construction, option parsing, the forward pass or the frozen readout constants. A unit-name resolution replaced upper-casing, and it is the identity for every country and profile unit. A check on the number of training pairs reads a

per-unit expectation that equals 526 for every unit outside the near-zero refits. Some functions gained annotations and parameters with backward-compatible defaults. The scoring loop gained a guard that stops with a named error when a scoring function is missing. The readout path is therefore not byte-identical across revisions, and what is verified is that every pin reproduces at the current revision and that the changes leave every scored unit's arithmetic unchanged.

The first-cohort scoring run. The first cohort's scores (surface S1) were persisted by a run whose manifest records a digest of the code bundle rather than a git commit, and none of the tracked source snapshots matches that digest. The commit that produced the run therefore cannot be named. Two properties of the run were verified directly. Its specification hash, which fixes the prompts, equals the strict pin, and the weights of the one adapter checked match the archived manifest byte for byte. A later replay of five selected items on the base model and that adapter, using the same four-bit quantized base model, lands within three units of half-precision rounding of the stored option log probabilities, which is the precision at which they were stored. The same replay on the unquantized model departs by far more. Quantization is therefore the source of the dominant difference between the two configurations, and the stored first-cohort scores are consistent with the quantized model that the adapters were trained on. The full panel was not replayed.

The archive of the development stage. Before every scoring run, a gate compared the base model's scores on the frozen software stack with the archived scores of the development stage, whose software environment was not pinned. The comparison is report-only by design. The formatted prompts matched the archive exactly, while the log probabilities differed by more than the tolerance used at the development stage. A replay with the same four-bit quantization moved the scores toward the archive without reproducing it, so the remaining cause of that gap is not established. The gap enters no statistic in the article, which uses only scores produced on the frozen stack.

Configuration. The configuration file `paper_b/config.py` serves both cohorts. Its first-cohort defaults are unchanged when the environment variable `SCA2_PB_UNITS` is unset, and the file is byte-identical to its version at the registration tag.

File hashes. Hashes of text files are given for the working-tree bytes. The item pool `pool_skeleton.csv` is written with Windows line endings by the pool-building script, while the repository stores text with Unix line endings, so the hash of the stored file differs from the hash of the file on disk. Both are given below. A reader who checks the freeze guard on a fresh checkout must regenerate the pool with `python paper_b/heldout/build_pool.py`, which reproduces the hashed bytes from the overrides file, because the guard compares raw bytes and refuses to run on the converted copy.

- Amended item pool, as written:
  `19241f6f103c51d64e295045162bb370920223b7ffad7c5f3c8991f5c101bd44`
- Amended item pool, as stored in the repository:
  `a1dbceea1a33da05f0432e14e0ff174ba239916b1a5beb9944e9aa0910cd40a5`
- Overrides file:
  `0f5cd2bbd88ef6eaca9cca9ded24e8a5e056276be055b21c1d92b941dd7d3849`
- Frozen primary and secondary selection:
  `782b7d2123f9758ea5641c56ddac6bf778fa4a01e3b6e7ce11402d224480bf21`